

Python For Computational Problem Solving- UE26CS151A

IDLE – Python Development Environment

Dr Manjula K
Professor, PCPS Anchor 2026
Department of CSE(AIML)

What is IDLE?

- IDLE stands for Integrated Development and Learning Environment.
- It is a simple Python development environment that comes with Python.
- It supports interactive programming and script development.
- It helps users write, run, test, and debug Python programs.

Why Use IDLE?

- Beginner-friendly interface
- Python Shell for interactive execution
- Code editor for writing programs
- Syntax highlighting
- Automatic indentation
- Built-in debugger
- Easy program execution

Installing IDLE

1. Download Python:
   - Go to the official Python Website and click on the download link
   - for Windows.
   - Save the executable file to your computer.
2. Run the Installer:
   - Double-click the downloaded file to start the setup window.
   - At the bottom of the installer window, make sure to check the box
   - that says "Add python.exe to PATH" (if available).
   - Click on "Install Now". This default setting installs Python along
   - with IDLE, pip, and documentation.

Installing IDLE

Finish Setup:

Wait for the progress bar to finish until a "Setup was successful" message appears.

How to Open IDLE

1. Press the **Windows Key** on your keyboard to open the Start menu.
2. Type IDLE in the search bar.
3. Click on **IDLE (Python 3.x)** to launch the application

Starting IDLE

```
cjc@Chiranth-2 ~ % python3
Python 3.14.4 (main, Apr  7 2026, 13:13:20) [Clang 21.0.0 (clang-2100.0.123.102)
] on darwin
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

When IDLE opens, the Python Shell is displayed.

The Shell provides the >>> prompt for entering Python statements.

IDLE Interface

- Menu Bar
- Python Shell
- Interactive prompt (>>>)
- Editor Window
- Output area
- Debugger and other development tools

Python Shell

The Python Shell executes Python statements interactively.





Interactive Mode

- Executes one statement at a time.
- Results appear immediately.
- Useful for learning, calculations, testing small code fragments, and experimenting with Python.

Script Mode

- Used to write complete Python programs.
- Programs are normally saved with the .py extension.
- Example: hello.py
- A complete program can be edited, saved, and executed repeatedly.

Creating a Python Program

- File → New File
- Write the Python program in the Editor window.
- File → Save
- Save using a meaningful .py filename.

Running a Program

- Use Run → Run Module.
- Keyboard shortcut: F5
- IDLE executes the current module.
- Output is displayed in the Python Shell.

Interactive Mode vs Script Mode

Interactive Mode	Script Mode
Executes one line at a time	Executes complete program
Immediate output	Saved as .py file
Good for testing	Good for projects

Script Mode Features

1: Syntax Highlighting

Syntax highlighting is a feature in IDLE that displays different parts of Python code in different colors, making the code easier to read and understand.

Benefits

- Highlights **keywords**, **strings**, **comments**, and other code elements.
- Improves code readability.
- Makes it easier to identify syntax mistakes.
- Helps beginners understand Python code more easily.

2: Automatic Indentation

- Python uses indentation to define blocks.
- IDLE assists with automatic indentation.
- Indentation is part of Python syntax.

```
>>> marks=50
>>> if marks>=40:
...     print("pass")
...
pass
>>>
```

Script Mode: Saving Python Programs

- Python source files use the .py extension.
- Examples: calculator.py, student.py, area.py
- Use File → Save or Ctrl + S.
- Use meaningful filenames.

Script Mode: Opening an Existing Program

- File → Open
- Select the required .py file.
- The file opens in the IDLE Editor Window.
- You can edit, save, and run it using F5.

Errors in IDLE

IDLE displays error messages when problems occur.

Error	Description	Example
SyntaxError	Invalid Python syntax	print(“Hello"
NameError	Variable not defined	print(x)
TypeError	Unsupported operation between data types	"5" + 5
ValueError	Invalid value passed to a function	int("abc")
IndentationError	Incorrect indentation in code	Missing spaces after if satement

Debugging in IDLE

Features of IDLE Debugger

- Built-in debugger included with IDLE.
- Executes programs step by step.
- Allows you to inspect variable values during execution.
- Helps identify logical and runtime errors.

Benefits

- Makes it easier to understand program flow.
- Simplifies locating and fixing bugs.
- Useful for testing and improving code.

Breakpoints

A breakpoint is a marker that temporarily pauses program execution at a selected line of code.

Why Use Breakpoints?

- Pause execution at a specific line.
- Examine the values of variables.
- Observe the flow of the program step by step.
- Help locate and fix logical errors.

Benefits

- Makes debugging easier.
- Improves understanding of program execution.
- Saves time when troubleshooting complex code.

Useful IDLE Shortcuts

Action	Windows	macOS
New File	Ctrl + N	⌘ + N
Open File	Ctrl + O	⌘ + O
Save File	Ctrl + S	⌘ + S
Run Module	F5	Fn + F5
Undo	Ctrl + Z	⌘ + Z
Find	Ctrl + F	⌘ + F
Select All	Ctrl + A	⌘ + A

Example: Area of a Circle

```
radius = float(input("Enter radius: "))
area = 3.14159 * radius * radius
print("Area =", area)
```

Example output: Enter radius: 5 → Area = 78.53975

IDLE Workflow

Create / Open File

↓

Write Program

↓

Save .py

↓

Run using F5

↓

View Output in Python Shell

Advantages of IDLE

- Comes with Python
- Easy to use
- Beginner-friendly
- Interactive Shell
- Built-in editor
- Syntax highlighting
- Automatic indentation
- Debugging support

Limitations of IDLE

- Basic interface
- Fewer advanced development features
- Limited project management
- Less suitable for very large projects
- Advanced users may prefer feature-rich IDEs such as VS Code or PyCharm.

Key Takeaways

- IDLE stands for Integrated Development and Learning Environment.
- It provides a Python Shell for interactive execution.
- The Editor Window is used to write and run Python scripts.
- Python programs are saved with the .py extension.
- Press F5 to run the current program.
- IDLE includes basic debugging tools to help find and fix errors.



Thank You

Department of CSE (AIML)

Dr Manjula K- manjulak@pes.edu

Ack: Teaching Assistant - Chiranth J(CSE AIML)